Supplementary Material for Integrated Port Energy System Experiments

I. Test Case Parameters and Operating Costs

Each period represents one hour. Monetary inputs and costs are in USD. Electricity and fuel prices are expressed per MWh of the corresponding carrier. MW H_2 and MWh H_2 denote hydrogen energy-equivalent rates and inventories. A dash denotes an inactive or inapplicable parameter.

TABLE I
Case configuration and indexing conventions

Setting	Case 1	Case 2	Case 3
Energy structure	Electricity–hydrogen	Electricity–heating–cooling	Electricity–hydrogen
Periods	t1–t24	t1–t72	t1–t24
Vessels / berths	16 / 6	22 / 6	11 / 5
Container tasks	6600	Not modelled	4022
QCs / YCs / ECTs / HCTs	30 / 40 / 25 / 10	Not modelled	12 / 20 / 25 / 15
Relative MIP gap	0%	0%	0.1%

Case 1 monetary inputs are converted at 6.7489 CNY/USD. Electricity sales are disabled in all three cases.

A. Energy device parameters

TABLE II
Energy device capacities efficiencies storage bounds and costs

Device/category	Parameter	Description	Unit	Case 1	Case 2	Case 3
Energy purchase	Pbmin	Minimum grid-purchase power	MW	0	0	0
Energy purchase	Pbmax	Maximum grid-purchase power	MW	30	120	20
Energy purchase	Hbmin	Minimum external hydrogen-purchase rate	MW H_2	0	—	0
Energy purchase	Hbmax	Maximum external hydrogen-purchase rate	MW H_2	30	—	20
Electrolyzer	Pedmin	Minimum electrical input	MW	0.25	—	1
Electrolyzer	Pedmax	Maximum electrical input	MW	2.5	—	5
Electrolyzer	RUed	Ramp-up limit	MW/h	2	—	3
Electrolyzer	RDed	Ramp-down limit	MW/h	2	—	3
Electrolyzer	SUed	Startup power limit	MW	1.5	—	2
Electrolyzer	SDed	Shutdown power limit	MW	1.5	—	2
Electrolyzer	Tedu	Minimum up time	h	2	—	4
Electrolyzer	tedd	Minimum down time	h	2	—	4
Electrolyzer	Ped0	Initial electrical input	MW	0	—	0
Electrolyzer	Xed0	Initial on/off state	binary	0	—	0
Electrolyzer	ned	Electricity-to-hydrogen efficiency	p.u.	0.7	—	0.7
Electrolyzer	Csu	Start-up cost	$/event	14.8172294744	—	30
Electrolyzer	Csd	Shut-down cost	$/event	14.8172294744	—	30
Fuel cell	Pfcmin	Minimum electrical output	MW	0	—	0
Fuel cell	Pfcmax	Maximum electrical output	MW	2.8	—	4
Fuel cell	nfc	Hydrogen-to-electricity efficiency	p.u.	0.5	—	0.6
Electric storage	Pescmin	Minimum charging power	MW	0	0	0
Electric storage	Pescmax	Maximum charging power	MW	2.5	12	5
Electric storage	Pesdmin	Minimum discharging power	MW	0	0	0
Electric storage	Pesdmax	Maximum discharging power	MW	2.5	12	5
Electric storage	Seesmin	Minimum stored energy	MWh	0.75	12	2
Electric storage	Seesmax	Maximum stored energy	MWh	3	40	20
Electric storage	Sees0	Initial stored energy	MWh	0.75	18	2

Device/category	Parameter	Description	Unit	Case 1	Case 2	Case 3
Electric storage	nesc	Charging efficiency	p.u.	0.95	0.85	0.95
Electric storage	nesd	Discharging efficiency	p.u.	0.95	0.98	0.95
Hydrogen storage	Hhscmin	Minimum charging rate	MW H_2	0	—	0
Hydrogen storage	Hhscmax	Maximum charging rate	MW H_2	2	—	3
Hydrogen storage	Hhsdmin	Minimum discharging rate	MW H_2	0	—	0
Hydrogen storage	Hhsdmax	Maximum discharging rate	MW H_2	2	—	3
Hydrogen storage	Shesmin	Minimum stored hydrogen energy	MWh H_2	3.3	—	1
Hydrogen storage	Shesmax	Maximum stored hydrogen energy	MWh H_2	13.3	—	10
Hydrogen storage	Shes0	Initial stored hydrogen energy	MWh H_2	3.3	—	1
Hydrogen storage	nhsc	Charging efficiency	p.u.	0.98	—	0.98
Hydrogen storage	nhsd	Discharging efficiency	p.u.	0.98	—	0.98
Renewables	Cresc	Renewable-curtailment penalty	$/MWh	74.0861473722	—	—
Gas turbine	Pgtmin	Minimum electrical output	MW	—	0	—
Gas turbine	Pgtmax	Maximum electrical output	MW	—	310	—
Gas turbine	RUgt	Ramp-up limit	MW/h	—	40	—
Gas turbine	RDgt	Ramp-down limit	MW/h	—	40	—
Gas turbine	ngte	Electrical efficiency	p.u.	—	0.3	—
Gas turbine	ngtt	Heat-recovery efficiency	p.u.	—	0.5	—
Heat exchanger	Thxmin	Minimum useful-heat output	MW	—	0	—
Heat exchanger	Thxmax	Maximum useful-heat output	MW	—	100	—
Heat exchanger	RUhx	Ramp-up limit	MW/h	—	20	—
Heat exchanger	RDhx	Ramp-down limit	MW/h	—	20	—
Heat exchanger	nhx	Conversion efficiency	p.u.	—	0.8	—
Electric boiler	Tebmin	Minimum heat output	MW	—	0	—
Electric boiler	Tebmax	Maximum heat output	MW	—	100	—
Electric boiler	RUeb	Ramp-up limit	MW/h	—	20	—
Electric boiler	RDeb	Ramp-down limit	MW/h	—	20	—
Electric boiler	neb	Conversion efficiency	p.u.	—	0.96	—
Absorption chiller	Cacmin	Minimum cooling output	MW	—	0	—
Absorption chiller	Cacmax	Maximum cooling output	MW	—	200	—
Absorption chiller	COPac	Coefficient of performance	p.u.	—	1.5	—
Electric refrigeration unit	Cermin	Minimum cooling output	MW	—	0	—
Electric refrigeration unit	Cermax	Maximum cooling output	MW	—	130	—
Electric refrigeration unit	COPer	Coefficient of performance	p.u.	—	3	—
Thermal storage	Ttescmin	Minimum charging power	MW	—	0	—
Thermal storage	Ttescmax	Maximum charging power	MW	—	10	—
Thermal storage	Ttesdmin	Minimum discharging power	MW	—	0	—
Thermal storage	Ttesdmax	Maximum discharging power	MW	—	10	—
Thermal storage	Stesmin	Minimum stored heat	MWh	—	9	—
Thermal storage	Stesmax	Maximum stored heat	MWh	—	30	—
Thermal storage	Stes0	Initial stored heat	MWh	—	12	—
Thermal storage	ntesc	Charging efficiency	p.u.	—	0.98	—
Thermal storage	ntesd	Discharging efficiency	p.u.	—	0.98	—

In Houston, natural gas uses an energy-equivalent basis with zetaNG = rhoNG = 1.

B. Logistics equipment parameters

TABLE III
CONTAINER-HANDLING AND TRUCK PARAMETERS FOR CASES 1 AND 3

Category	Parameter	Description	Unit	Case 1	Case 3
Vessel scheduling	Cpea	Vessel waiting penalty	$/h	29.6344589489	15
Vessel scheduling	Cped	Vessel departure-delay penalty	$/h	29.6344589489	15
Handling capacity	Nqcmax	QC handling capacity	container/(unit·h)	30	40
Handling capacity	Nycmax	YC handling capacity	container/(unit·h)	35	33
Handling capacity	Nectmax	ECT transport capacity	container/(unit·h)	20	42
Handling capacity	Nhctmax	HCT transport capacity	container/(unit·h)	40	45
QC allocation	nqca	Maximum QCs assigned to one vessel	unit	Vessel-specific	4
Yard state	Nymin	Minimum yard inventory state	container	0	—
Yard state	Nymax	Maximum yard inventory state	container	240	—
Yard state	Ny0	Initial yard inventory state	container	0	—
Handling energy	pqcu	QC electricity coefficient	see note	0.32	0.006
Handling energy	pycu	YC electricity coefficient	see note	0.35	0.0045
Handling energy	pectu	ECT electricity coefficient	see note	0.04	0.004
Handling energy	hhctu	HCT hydrogen coefficient	see note	0.04	0.003
ECT	Pectcmin	Minimum charging power per ECT	MW	0	0
ECT	Pectcmax	Maximum charging power per ECT	MW	0.16	0.05
ECT	Sectmin	Minimum battery energy per ECT	MWh	0.07	0.08
ECT	Sectmax	Maximum battery energy per ECT	MWh	0.35	0.36
ECT	Sect0	Initial battery energy per ECT	MWh	0.07	0.08
ECT	nectc	ECT charging efficiency	p.u.	0.95	0.95
ECT	nectd	ECT discharging efficiency	p.u.	0.95	0.95
HCT	Hhctcmin	Minimum hydrogen-refueling rate per HCT	MW H_2	0	0
HCT	Hhctcmax	Maximum hydrogen-refueling rate per HCT	MW H_2	0.3	0.035
HCT	Shctmin	Minimum onboard hydrogen energy per HCT	MWh H_2	0.1	0.045
HCT	Shctmax	Maximum onboard hydrogen energy per HCT	MWh H_2	0.6	0.285
HCT	Shct0	Initial onboard hydrogen energy per HCT	MWh H_2	0.1	0.045
HCT	nhctc	HCT refueling efficiency	p.u.	0.98	0.98
HCT	nhctd	HCT discharge efficiency	p.u.	0.98	0.98

Case 1 coefficients are MW or MW H_2 per active unit; Case 3 coefficients are MWh or MWh H_2 per container. Case 1 uses vessel-specific QC bounds, while Case 3 allows at most four QCs per vessel.

Case 1 uses fixed-throughput handling and terminal electric, hydrogen and truck inventories no lower than their initial values. Case 3 uses per-container throughput and terminal inventories equal to their initial values. Houston uses fixed-duration vessel service, equal initial and terminal storage states, and zero vessel waiting penalties.

C. Vessel parameters

TABLE IV
ZHOUSHAN VESSEL WINDOWS TASKS SHORE-POWER DEMANDS AND QC BOUNDS

Vessel	Earliest arrival (h)	Latest departure (h)	Container task	Shore power (MW)	Minimum QCs	Maximum QCs
v1	1	8	400	1.6	1	4
v2	3	11	350	1.4	2	3
v3	4	11	550	1.9	1	4
v4	7	15	450	1.8	2	4
v5	8	16	300	1.2	2	3
v6	8	17	400	1.5	1	3
v7	9	17	450	1.6	1	3
v8	12	18	400	1.7	1	3
v9	13	20	450	1.5	1	4
v10	13	20	400	1.4	1	4
v11	14	21	550	2.2	1	4
v12	14	22	450	1.7	1	4
v13	15	24	400	1.6	1	3
v14	17	23	300	1.3	1	3
v15	18	24	350	1.5	2	4
v16	19	24	400	1.4	2	5

TABLE V
HOUSTON VESSEL ARRIVALS SERVICE DURATIONS AND MEAN ENERGY DEMANDS

Vessel	Reported arrival (h)	Required berthing (h)	Type	Size code	Electricity (MW)	Heating (MW)	Cooling (MW)
v1	1	20	Cruise	2	10	30	0
v2	1	36	Cruise	3	20	50	0
v3	4	8	Cruise	1	4	10	0
v4	6	20	Cruise	2	10	30	0
v5	9	20	Cruise	2	10	30	0
v6	13	8	Cruise	1	4	10	0
v7	15	8	Cruise	1	4	10	0
v8	21	20	Cruise	2	10	30	0
v9	26	8	Cruise	1	4	10	0
v10	34	8	Cruise	1	4	10	0
v11	45	8	Cruise	1	4	10	0
v12	1	36	Reefer	3	20	0	50
v13	3	8	Reefer	1	4	0	10
v14	3	20	Reefer	2	10	0	30
v15	5	8	Reefer	1	4	0	10
v16	10	36	Reefer	3	20	0	50
v17	16	20	Reefer	2	10	0	30
v18	20	8	Reefer	1	4	0	10
v19	23	20	Reefer	2	10	0	30
v20	30	8	Reefer	1	4	0	10
v21	36	8	Reefer	1	4	0	10
v22	54	8	Reefer	1	4	0	10

TABLE VI
YANGSHAN VESSEL WINDOWS CONTAINER TASKS AND SHORE-POWER DEMANDS

Vessel	Earliest arrival (h)	Latest departure (h)	Container task	Shore power (MW)
v1	3	8	259	0.7
v2	4	10	356	0.7
v3	8	14	435	0.7
v4	9	15	428	1.99
v5	10	15	333	0.63
v6	10	16	455	0.7
v7	11	16	350	0.43
v8	12	20	684	1.26
v9	18	23	400	0.83
v10	19	23	150	0.32
v11	20	24	172	0.32

In Table V, size codes 1, 2 and 3 denote small, medium and large vessels, respectively. Houston's demands combine vessel means, hourly factors and berth occupancy; fixed base loads are zero.

$$L(c,t) = f(c,t) \times \Sigma(v)\,[P(c,v)\,X(v,t)]$$

Here c denotes electricity, heating or cooling; $\bar{P}$(c,v) is the corresponding vessel mean in Table V, f(c,t) is its time-series factor, and X(v,t) indicates whether vessel v occupies a berth in period t.

D. Time series parameters

Tables VII and VIII give the Zhoushan and Yangshan hourly inputs. Renewable availability is in MW, with resource IDs i1–i3 for Case 1 and res for Case 3. The hourly energy prices, fixed loads, and available renewable power for the Houston case are provided in reference [5] of the main paper.

TABLE VII
ZHOUSHAN HOURLY PRICES FIXED DEMAND AND RENEWABLE AVAILABILITY

Period	Electricity USD/MWh	Hydrogen USD/MWh H_2	Fixed electric load MW	i1 MW	i2 MW	i3 MW
t1	32.9683355806	66.6775326349	14.88	1.502	1.502	0
t2	32.9683355806	66.6775326349	14.742	1.291	1.291	0
t3	32.9683355806	66.6775326349	13.593	0.703	0.703	0
t4	32.9683355806	66.6775326349	14.536	2.463	2.463	0
t5	32.9683355806	66.6775326349	14.3	3.436	3.436	0
t6	32.9683355806	66.6775326349	12.119	3.009	3.009	0
t7	32.9683355806	66.6775326349	13.028	3.374	3.374	0.063
t8	32.9683355806	66.6775326349	11.907	3.346	3.346	0.369
t9	143.149253952	66.6775326349	10.732	3.881	3.881	0.695
t10	143.149253952	66.6775326349	12.72	6.25	6.25	0.816
t11	143.149253952	66.6775326349	12.9	4.692	4.692	1.414
t12	32.9683355806	66.6775326349	11.971	5.686	5.685	1.52
t13	32.9683355806	66.6775326349	11.158	6.239	6.239	1.464
t14	143.149253952	66.6775326349	11.072	6.225	6.225	1.159
t15	143.149253952	66.6775326349	10.645	6.128	6.128	0.605
t16	143.149253952	66.6775326349	10.598	5.34	5.34	0.527
t17	143.149253952	66.6775326349	9.992	4.477	4.478	0.216
t18	86.7548785728	66.6775326349	9.675	5.788	5.788	0.033
t19	86.7548785728	66.6775326349	11.048	5.424	5.424	0
t20	86.7548785728	66.6775326349	11.138	4.494	4.494	0
t21	86.7548785728	66.6775326349	11.505	3.54	3.541	0
t22	86.7548785728	66.6775326349	11.262	5.748	5.748	0
t23	86.7548785728	66.6775326349	10.53	4.509	4.509	0
t24	32.9683355806	66.6775326349	10.545	3.321	3.321	0

TABLE VIII
YANGSHAN HOURLY PRICES FIXED DEMAND AND RENEWABLE AVAILABILITY

Period	Electricity USD/MWh	Hydrogen USD/MWh H_2	Fixed electric load MW	res MW
t1	47	50	2.3	1.7
t2	24	50	2	1.8
t3	24	50	1.9	1.9
t4	24	50	1.8	2.4
t5	33	50	1.8	2.3
t6	42	50	1.8	1.9
t7	49	50	2	2
t8	76	50	2.1	2.1
t9	115	50	2.2	1.9
t10	134	50	2.3	1.5
t11	135	50	3.7	1.1
t12	123	50	4.2	1.2
t13	81	50	4	1.2
t14	82	50	4.2	1.4
t15	92	50	4.3	2.1
t16	91	50	4.2	2.2
t17	91	50	4.4	2.3
t18	91	50	4.6	2.2
t19	134	50	4.8	2.1
t20	100	50	4.2	1.9
t21	75	50	3.8	1.7
t22	67	50	3.3	2.1
t23	44	50	3	2.1
t24	43	50	2.7	2.1

E. Baseline operating costs

Costs are in USD. Inactive components contribute zero.

TABLE IX
BASELINE OPERATING COST BREAKDOWN

Cost component	**Case 1**	**Case 2**	**Case 3**
Electricity purchases	18,117.30	38,978.87	4,110.17
Grid-sale credit	0.00	0.00	0.00
Hydrogen purchases	2,109.82	0.00	2,591.31
Natural gas purchases	0.00	82,631.85	0.00
ED startup	29.63	0.00	30.00
ED shutdown	29.63	0.00	30.00
RES curtailment penalty	0.00	0.00	0.00
Vessel waiting penalty	355.61	0.00	75.00
Vessel departure-delay penalty	0.00	0.00	0.00
Total operating cost	**20,642.00**	**121,610.72**	**6,836.48**

Values are rounded to two decimals; totals use unrounded values.

II. Scenario Analysis Tasks

Each port has 100 tasks: 30 energy-parameter adjustments (E01–E30), 30 equipment-constraint adjustments (Q01–Q30), 30 logistics adjustments (L01–L30) and 10 energy–logistics coupled adjustments (X01–X10). Prefixes Z, H and Y denote Zhoushan, Houston and Yangshan, respectively.

Each task changes only the specified inputs or adds the specified constraints. Multipliers apply to the original values, and all period ranges are inclusive. Coupled modifications form one scenario.

A bound on a sum is one aggregate constraint. Changes to equipment counts retain the capacity of each unit.

TABLE X
INDEXED VARIABLES USED IN THE TASK CATALOGUE

Symbol	Meaning	Unit
Pb	Grid-purchase power	MW
Pesc / Pesd / Sees	Electric storage charging / discharging / inventory	MW / MW / MWh
Ped / Pfc	Electrolyzer electrical input / fuel-cell electrical output	MW
Hb / Hhsc / Hhsd / Shes	Hydrogen purchases / storage charging / discharging / inventory	MW H_2 / MW H_2 / MW H_2 / MWh H_2
Pgt / Thx / Teb	Gas-turbine electric output / exchanger heat output / boiler heat output	MW
Cac / Cer	Absorption chiller / electric refrigeration unit cooling output	MW
Ttesc / Ttesd / Stes	Thermal storage charging / discharging / inventory	MW / MW / MWh

A. Zhoushan Port

TABLE XI
ZHOUSHAN PORT ENERGY PARAMETER ADJUSTMENTS

Task	*Requested modifications*
Z-E01	Multiply grid electricity purchase tariff at t1 by 1.08.
Z-E02	Multiply grid electricity purchase tariff at t4–t6 by 0.93.
Z-E03	Multiply grid electricity purchase tariff at t2, t9, t16 by 1.12.
Z-E04	Multiply grid electricity purchase tariff at t23, t24 by 0.94.
Z-E05	Multiply grid electricity purchase tariff at t1–t24 by 1.04.
Z-E06	Multiply grid electricity purchase tariff at t1–t12 by 1.06.
Z-E07	Multiply grid electricity purchase tariff at t13–t24 by 0.92.
Z-E08	Multiply grid electricity purchase tariff at t3, t4, t15, t16 by 1.09.
Z-E09	Multiply external hydrogen purchase price at t5, t6 by 1.07.
Z-E10	Multiply external hydrogen purchase price at t2, t12, t22 by 0.94.
Z-E11	Multiply external hydrogen purchase price at t1–t24 by 1.03.
Z-E12	Multiply external hydrogen purchase price at t22–t24 by 1.11.
Z-E13	Multiply available power of i1, i2, i3 at t7–t9 by 0.94.
Z-E14	Multiply available power of i1 at t11–t13 by 1.06.
Z-E15	Multiply available power of i3 at t14–t16 by 0.92.
Z-E16	Multiply available power of i1, i2, i3 at t6, t12, t18 by 0.96.
Z-E17	Multiply available power of i1, i2, i3 at t8–t11 by 1.08.
Z-E18	Multiply available power of i1 at t1–t24 by 0.97.
Z-E19	Multiply available power of i3 at t7, t8, t16, t17 by 0.93.
Z-E20	Multiply available power of i1, i2, i3 at t12 by 1.12.

Task	*Requested modifications*
Z-E21	Multiply maximum electric-storage inventory (Seesmax) by 1.12.
Z-E22	Multiply minimum electric-storage inventory (Seesmin) by 0.92.
Z-E23	Multiply electric-storage charging efficiency (nesc) by 0.98.
Z-E24	Multiply electric-storage discharging efficiency (nesd) by 0.97.
Z-E25	Multiply maximum hydrogen-storage inventory (Shesmax) by 1.13.
Z-E26	Multiply hydrogen-storage charging efficiency (nhsc) by 0.98.
Z-E27	Multiply hydrogen-storage discharging efficiency (nhsd) by 0.97.
Z-E28	Multiply electrolyzer conversion efficiency (ned) by 0.98.
Z-E29	Multiply fuel-cell electrical efficiency (nfc) by 0.97.
Z-E30	Multiply initial electric-storage inventory (Sees0) by 1.08.

TABLE XII
ZHOUSHAN PORT EQUIPMENT CONSTRAINT ADJUSTMENTS

Task	*Requested modifications*
Z-Q01	Add Pb[t5] ≤ 28.8 MW.
Z-Q02	Add Pesc[t6] ≤ 2.4 MW.
Z-Q03	Add Pesd[t7] ≤ 2.4 MW.
Z-Q04	Add Sees[t8] ≤ 2.88 MWh.
Z-Q05	Add Ped[t9] ≤ 2.4 MW.
Z-Q06	Add Pfc[t10] ≤ 2.688 MW.
Z-Q07	Add Hb[t11] ≤ 28.8 MW H_2.
Z-Q08	Add Hhsc[t12] ≤ 1.92 MW H_2.
Z-Q09	Add Hhsd[t13] ≤ 1.92 MW H_2.
Z-Q10	Add Shes[t14] ≤ 12.768 MWh H_2.
Z-Q11	Add Pb[t2] ≤ 28.2 MW; Add Pb[t3] ≤ 28.2 MW; Add Pb[t4] ≤ 28.2 MW.
Z-Q12	Add Pesc[t6] ≤ 2.35 MW; Add Pesc[t13] ≤ 2.35 MW; Add Pesc[t20] ≤ 2.35 MW.
Z-Q13	Add Pesd[t21] ≤ 2.35 MW; Add Pesd[t22] ≤ 2.35 MW; Add Pesd[t23] ≤ 2.35 MW.
Z-Q14	Add Sees[t2] ≤ 2.82 MWh; Add Sees[t3] ≤ 2.82 MWh; Add Sees[t4] ≤ 2.82 MWh.
Z-Q15	Add Ped[t6] ≤ 2.35 MW; Add Ped[t13] ≤ 2.35 MW; Add Ped[t20] ≤ 2.35 MW.
Z-Q16	Add Pfc[t21] ≤ 2.632 MW; Add Pfc[t22] ≤ 2.632 MW; Add Pfc[t23] ≤ 2.632 MW.
Z-Q17	Add Hb[t2] ≤ 28.2 MW H_2; Add Hb[t3] ≤ 28.2 MW H_2; Add Hb[t4] ≤ 28.2 MW H_2.
Z-Q18	Add Hhsc[t6] ≤ 1.88 MW H_2; Add Hhsc[t13] ≤ 1.88 MW H_2; Add Hhsc[t20] ≤ 1.88 MW H_2.
Z-Q19	Add Hhsd[t21] ≤ 1.88 MW H_2; Add Hhsd[t22] ≤ 1.88 MW H_2; Add Hhsd[t23] ≤ 1.88 MW H_2.
Z-Q20	Add Shes[t2] ≤ 12.502 MWh H_2; Add Shes[t3] ≤ 12.502 MWh H_2; Add Shes[t4] ≤ 12.502 MWh H_2.
Z-Q21	Add Pb[t8] ≤ 29.1 MW; Add Pesc[t8] ≤ 2.425 MW.
Z-Q22	Add Pesd[t9] ≤ 2.425 MW; Add Sees[t9] ≤ 2.91 MWh.
Z-Q23	Add Ped[t10] ≤ 2.425 MW; Add Pfc[t10] ≤ 2.716 MW.
Z-Q24	Add Hb[t11] ≤ 29.1 MW H_2; Add Hhsc[t11] ≤ 1.94 MW H_2.
Z-Q25	Add Hhsd[t12] ≤ 1.94 MW H_2; Add Shes[t12] ≤ 12.901 MWh H_2.

Task	*Requested modifications*
Z-Q26	Add Pb[t9] + Pb[t10] ≤ 55.5 MW.
Z-Q27	Add Pesc[t7] + Pesc[t19] ≤ 4.625 MW.
Z-Q28	Add Pesd[t9] + Pesd[t10] ≤ 4.625 MW.
Z-Q29	Add Sees[t7] + Sees[t19] ≤ 5.55 MWh.
Z-Q30	Add Ped[t9] + Ped[t10] ≤ 4.625 MW.

TABLE XIII
PORT LOGISTICS ADJUSTMENTS FOR ZHOUSHAN PORT

Task	*Requested modifications*
Z-L01	Increase vessel v2's earliest arrival by 2 h.
Z-L02	Decrease vessel v2's earliest arrival by 1 h.
Z-L03	Increase vessel v2's container workload by 20 containers.
Z-L04	Multiply vessel v2's shore-power demand by 0.92.
Z-L05	Increase vessel v2's latest departure by 2 h.
Z-L06	Increase earliest arrival for vessels v2 and v5 by 1 h.
Z-L07	Multiply vessel v2's shore-power demand by 1.07; Multiply vessel v5's shore-power demand by 0.95.
Z-L08	Increase vessel v2's earliest arrival by 1 h; Increase vessel v2's container workload by 15 containers.
Z-L09	Increase vessel v2's earliest arrival by 1 h; Multiply vessel v5's shore-power demand by 1.06.
Z-L10	Increase available QC count (NqcCount) by 1 unit; Increase vessel v2's container workload by 10 containers.
Z-L11	Increase vessel v3's earliest arrival by 2 h.
Z-L12	Decrease vessel v3's earliest arrival by 1 h.
Z-L13	Increase vessel v3's container workload by 20 containers.
Z-L14	Multiply vessel v3's shore-power demand by 0.92.
Z-L15	Increase vessel v3's latest departure by 2 h.
Z-L16	Increase earliest arrival for vessels v3 and v6 by 1 h.
Z-L17	Multiply vessel v3's shore-power demand by 1.07; Multiply vessel v6's shore-power demand by 0.95.
Z-L18	Increase vessel v3's earliest arrival by 1 h; Increase vessel v3's container workload by 15 containers.
Z-L19	Increase vessel v3's earliest arrival by 1 h; Multiply vessel v6's shore-power demand by 1.06.
Z-L20	Increase available YC count (NycCount) by 1 unit; Increase vessel v3's container workload by 10 containers.
Z-L21	Increase vessel v4's earliest arrival by 2 h.
Z-L22	Decrease vessel v4's earliest arrival by 1 h.
Z-L23	Increase vessel v4's container workload by 20 containers.
Z-L24	Multiply vessel v4's shore-power demand by 0.92.
Z-L25	Increase vessel v4's latest departure by 2 h.
Z-L26	Increase earliest arrival for vessels v4 and v7 by 1 h.
Z-L27	Multiply vessel v4's shore-power demand by 1.07; Multiply vessel v7's shore-power demand by 0.95.
Z-L28	Increase vessel v4's earliest arrival by 1 h; Increase vessel v4's container workload by 15 containers.
Z-L29	Increase vessel v4's earliest arrival by 1 h; Multiply vessel v7's shore-power demand by 1.06.
Z-L30	Increase available ECT count (NectCount) by 1 unit; Increase vessel v4's container workload by 10 containers.

TABLE XIV
ZHOUSHAN PORT ENERGY LOGISTICS COUPLED ADJUSTMENTS

Task	*Requested modifications*
Z-X01	Multiply grid electricity purchase tariff at t4, t5 by 1.07; Increase vessel v2's earliest arrival by 1 h.
Z-X02	Multiply grid electricity purchase tariff at t5, t6 by 1.07; Increase vessel v3's earliest arrival by 1 h.
Z-X03	Multiply grid electricity purchase tariff at t6, t7 by 1.07; Increase vessel v4's earliest arrival by 1 h.
Z-X04	Multiply available power of i1, i2, i3 at t7, t8 by 0.95; Multiply vessel v2's shore-power demand by 1.05.
Z-X05	Multiply available power of i1, i2, i3 at t8, t9 by 0.95; Multiply vessel v3's shore-power demand by 1.05.
Z-X06	Multiply available power of i1, i2, i3 at t9, t10 by 0.95; Multiply vessel v4's shore-power demand by 1.05.
Z-X07	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v2's container workload by 10 containers.
Z-X08	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v3's container workload by 10 containers.
Z-X09	Multiply external hydrogen purchase price at t12, t13 by 1.06; Increase vessel v4's earliest arrival by 1 h; Multiply vessel v4's shore-power demand by 1.03.
Z-X10	Multiply external hydrogen purchase price at t13, t14 by 1.06; Increase vessel v2's earliest arrival by 1 h; Multiply vessel v2's shore-power demand by 1.03.

B. Port of Houston

TABLE XV
PORT OF HOUSTON ENERGY PARAMETER ADJUSTMENTS

Task	*Requested modifications*
H-E01	Multiply grid electricity purchase tariff at t1 by 1.08.
H-E02	Multiply grid electricity purchase tariff at t4–t6 by 0.93.
H-E03	Multiply grid electricity purchase tariff at t2, t9, t16 by 1.12.
H-E04	Multiply grid electricity purchase tariff at t71, t72 by 0.94.
H-E05	Multiply grid electricity purchase tariff at t1–t72 by 1.04.
H-E06	Multiply grid electricity purchase tariff at t1–t36 by 1.06.
H-E07	Multiply grid electricity purchase tariff at t37–t72 by 0.92.
H-E08	Multiply grid electricity purchase tariff at t3, t4, t15, t16 by 1.09.
H-E09	Multiply natural-gas purchase price at t5, t6 by 1.07.
H-E10	Multiply natural-gas purchase price at t2, t12, t22 by 0.94.
H-E11	Multiply natural-gas purchase price at t1–t72 by 1.03.
H-E12	Multiply natural-gas purchase price at t70–t72 by 1.11.
H-E13	Multiply available power of wind, pv at t7–t9 by 0.94.
H-E14	Multiply available power of wind at t11–t13 by 1.06.
H-E15	Multiply available power of pv at t14–t16 by 0.92.
H-E16	Multiply available power of wind, pv at t6, t12, t18 by 0.96.
H-E17	Multiply available power of wind, pv at t8–t11 by 1.08.
H-E18	Multiply available power of wind at t1–t72 by 0.97.
H-E19	Multiply available power of pv at t7, t8, t16, t17 by 0.93.
H-E20	Multiply available power of wind, pv at t12 by 1.12.

Task	*Requested modifications*
H-E21	Multiply maximum electric-storage inventory (Seesmax) by 1.12.
H-E22	Multiply minimum electric-storage inventory (Seesmin) by 0.92.
H-E23	Multiply electric-storage charging efficiency (nesc) by 0.98.
H-E24	Multiply electric-storage discharging efficiency (nesd) by 0.97.
H-E25	Multiply maximum thermal-storage inventory (Stesmax) by 1.13.
H-E26	Multiply thermal-storage charging efficiency (ntesc) by 0.98.
H-E27	Multiply thermal-storage discharging efficiency (ntesd) by 0.97.
H-E28	Multiply gas-turbine electrical efficiency (ngte) by 0.98.
H-E29	Multiply gas-turbine heat-recovery efficiency (ngtt) by 0.97.
H-E30	Multiply initial electric-storage inventory (Sees0) by 1.08.

TABLE XVI
PORT OF HOUSTON EQUIPMENT CONSTRAINT ADJUSTMENTS

Task	*Requested modifications*
H-Q01	Add Pb[t5] ≤ 115.2 MW.
H-Q02	Add Pesc[t6] ≤ 11.52 MW.
H-Q03	Add Pesd[t7] ≤ 11.52 MW.
H-Q04	Add Sees[t8] ≤ 38.4 MWh.
H-Q05	Add Pgt[t9] ≤ 297.6 MW.
H-Q06	Add Teb[t10] ≤ 96 MW.
H-Q07	Add Thx[t11] ≤ 96 MW.
H-Q08	Add Cer[t12] ≤ 124.8 MW.
H-Q09	Add Cac[t13] ≤ 192 MW.
H-Q10	Add Stes[t14] ≤ 28.8 MWh.
H-Q11	Add Pb[t2] ≤ 112.8 MW; Add Pb[t3] ≤ 112.8 MW; Add Pb[t4] ≤ 112.8 MW.
H-Q12	Add Pesc[t6] ≤ 11.28 MW; Add Pesc[t13] ≤ 11.28 MW; Add Pesc[t20] ≤ 11.28 MW.
H-Q13	Add Pesd[t69] ≤ 11.28 MW; Add Pesd[t70] ≤ 11.28 MW; Add Pesd[t71] ≤ 11.28 MW.
H-Q14	Add Sees[t2] ≤ 37.6 MWh; Add Sees[t3] ≤ 37.6 MWh; Add Sees[t4] ≤ 37.6 MWh.
H-Q15	Add Pgt[t6] ≤ 291.4 MW; Add Pgt[t13] ≤ 291.4 MW; Add Pgt[t20] ≤ 291.4 MW.
H-Q16	Add Teb[t69] ≤ 94 MW; Add Teb[t70] ≤ 94 MW; Add Teb[t71] ≤ 94 MW.
H-Q17	Add Thx[t2] ≤ 94 MW; Add Thx[t3] ≤ 94 MW; Add Thx[t4] ≤ 94 MW.
H-Q18	Add Cer[t6] ≤ 122.2 MW; Add Cer[t13] ≤ 122.2 MW; Add Cer[t20] ≤ 122.2 MW.
H-Q19	Add Cac[t69] ≤ 188 MW; Add Cac[t70] ≤ 188 MW; Add Cac[t71] ≤ 188 MW.
H-Q20	Add Stes[t2] ≤ 28.2 MWh; Add Stes[t3] ≤ 28.2 MWh; Add Stes[t4] ≤ 28.2 MWh.
H-Q21	Add Pb[t8] ≤ 116.4 MW; Add Pesc[t8] ≤ 11.64 MW.
H-Q22	Add Pesd[t9] ≤ 11.64 MW; Add Sees[t9] ≤ 38.8 MWh.
H-Q23	Add Pgt[t10] ≤ 300.7 MW; Add Teb[t10] ≤ 97 MW.
H-Q24	Add Thx[t11] ≤ 97 MW; Add Cer[t11] ≤ 126.1 MW.
H-Q25	Add Cac[t12] ≤ 194 MW; Add Stes[t12] ≤ 29.1 MWh.

Task	*Requested modifications*
H-Q26	Add Pb[t9] + Pb[t10] ≤ 222 MW.
H-Q27	Add Pesc[t7] + Pesc[t19] ≤ 22.2 MW.
H-Q28	Add Pesd[t9] + Pesd[t10] ≤ 22.2 MW.
H-Q29	Add Sees[t7] + Sees[t19] ≤ 74 MWh.
H-Q30	Add Pgt[t9] + Pgt[t10] ≤ 573.5 MW.

TABLE XVII
PORT LOGISTICS ADJUSTMENTS FOR THE PORT OF HOUSTON

Task	*Requested modifications*
H-L01	Increase vessel v2's reported arrival by 2 h.
H-L02	Increase vessel v2's reported arrival by 3 h.
H-L03	Increase vessel v2's required berthing duration by 2 h.
H-L04	Multiply vessel v2's mean electrical demand by 0.92.
H-L05	Multiply vessel v2's mean heating demand by 0.94.
H-L06	Increase reported arrival for vessels v2 and v5 by 1 h.
H-L07	Multiply vessel v2's mean electrical demand by 1.07; Multiply vessel v5's mean electrical demand by 0.95.
H-L08	Increase vessel v2's reported arrival by 1 h; Increase vessel v2's required berthing duration by 1 h.
H-L09	Increase vessel v2's reported arrival by 1 h; Multiply vessel v5's mean electrical demand by 1.06.
H-L10	Increase vessel v2's required berthing duration by 1 h; Multiply vessel v5's mean electrical demand by 1.04.
H-L11	Increase vessel v3's reported arrival by 2 h.
H-L12	Decrease vessel v3's reported arrival by 1 h.
H-L13	Increase vessel v3's required berthing duration by 2 h.
H-L14	Multiply vessel v3's mean electrical demand by 0.92.
H-L15	Multiply vessel v3's mean heating demand by 0.94.
H-L16	Increase reported arrival for vessels v3 and v6 by 1 h.
H-L17	Multiply vessel v3's mean electrical demand by 1.07; Multiply vessel v6's mean electrical demand by 0.95.
H-L18	Increase vessel v3's reported arrival by 1 h; Increase vessel v3's required berthing duration by 1 h.
H-L19	Increase vessel v3's reported arrival by 1 h; Multiply vessel v6's mean electrical demand by 1.06.
H-L20	Increase vessel v3's required berthing duration by 1 h; Multiply vessel v6's mean electrical demand by 1.04.
H-L21	Increase vessel v4's reported arrival by 2 h.
H-L22	Decrease vessel v4's reported arrival by 1 h.
H-L23	Increase vessel v4's required berthing duration by 2 h.
H-L24	Multiply vessel v4's mean electrical demand by 0.92.
H-L25	Multiply vessel v4's mean heating demand by 0.94.
H-L26	Increase reported arrival for vessels v4 and v7 by 1 h.
H-L27	Multiply vessel v4's mean electrical demand by 1.07; Multiply vessel v7's mean electrical demand by 0.95.
H-L28	Increase vessel v4's reported arrival by 1 h; Increase vessel v4's required berthing duration by 1 h.
H-L29	Increase vessel v4's reported arrival by 1 h; Multiply vessel v7's mean electrical demand by 1.06.
H-L30	Increase vessel v4's required berthing duration by 1 h; Multiply vessel v7's mean electrical demand by 1.04.

TABLE XVIII
PORT OF HOUSTON ENERGY LOGISTICS COUPLED ADJUSTMENTS

Task	*Requested modifications*
H-X01	Multiply grid electricity purchase tariff at t4, t5 by 1.07; Increase vessel v2's reported arrival by 1 h.
H-X02	Multiply grid electricity purchase tariff at t5, t6 by 1.07; Increase vessel v3's reported arrival by 1 h.
H-X03	Multiply grid electricity purchase tariff at t6, t7 by 1.07; Increase vessel v4's reported arrival by 1 h.
H-X04	Multiply available power of wind, pv at t7, t8 by 0.95; Multiply vessel v2's mean electrical demand by 1.05.
H-X05	Multiply available power of wind, pv at t8, t9 by 0.95; Multiply vessel v3's mean electrical demand by 1.05.
H-X06	Multiply available power of wind, pv at t9, t10 by 0.95; Multiply vessel v4's mean electrical demand by 1.05.
H-X07	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v2's required berthing duration by 1 h.
H-X08	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v3's required berthing duration by 1 h.
H-X09	Multiply natural-gas purchase price at t12, t13 by 1.06; Increase vessel v4's reported arrival by 1 h; Multiply vessel v4's mean electrical demand by 1.03.
H-X10	Multiply natural-gas purchase price at t13, t14 by 1.06; Increase vessel v2's reported arrival by 1 h; Multiply vessel v2's mean electrical demand by 1.03.

C. Yangshan Port

TABLE XIX
YANGSHAN PORT ENERGY PARAMETER ADJUSTMENTS

Task	*Requested modifications*
Y-E01	Multiply grid electricity purchase tariff at t1 by 1.08.
Y-E02	Multiply grid electricity purchase tariff at t4–t6 by 0.93.
Y-E03	Multiply grid electricity purchase tariff at t2, t9, t16 by 1.12.
Y-E04	Multiply grid electricity purchase tariff at t23, t24 by 0.94.
Y-E05	Multiply grid electricity purchase tariff at t1–t24 by 1.04.
Y-E06	Multiply grid electricity purchase tariff at t1–t12 by 1.06.
Y-E07	Multiply grid electricity purchase tariff at t13–t24 by 0.92.
Y-E08	Multiply grid electricity purchase tariff at t3, t4, t15, t16 by 1.09.
Y-E09	Multiply external hydrogen purchase price at t5, t6 by 1.07.
Y-E10	Multiply external hydrogen purchase price at t2, t12, t22 by 0.94.
Y-E11	Multiply external hydrogen purchase price at t1–t24 by 1.03.
Y-E12	Multiply external hydrogen purchase price at t22–t24 by 1.11.
Y-E13	Multiply available power of res at t7–t9 by 0.94.
Y-E14	Multiply available power of res at t11–t13 by 1.06.
Y-E15	Multiply available power of res at t14–t16 by 0.92.
Y-E16	Multiply available power of res at t6, t12, t18 by 0.96.
Y-E17	Multiply available power of res at t8–t11 by 1.08.
Y-E18	Multiply available power of res at t1–t24 by 0.97.
Y-E19	Multiply available power of res at t7, t8, t16, t17 by 0.93.
Y-E20	Multiply available power of res at t12 by 1.12.

Task	*Requested modifications*
Y-E21	Multiply maximum electric-storage inventory (Seesmax) by 1.12.
Y-E22	Multiply minimum electric-storage inventory (Seesmin) by 0.92.
Y-E23	Multiply electric-storage charging efficiency (nesc) by 0.98.
Y-E24	Multiply electric-storage discharging efficiency (nesd) by 0.97.
Y-E25	Multiply maximum hydrogen-storage inventory (Shesmax) by 1.13.
Y-E26	Multiply hydrogen-storage charging efficiency (nhsc) by 0.98.
Y-E27	Multiply hydrogen-storage discharging efficiency (nhsd) by 0.97.
Y-E28	Multiply electrolyzer conversion efficiency (ned) by 0.98.
Y-E29	Multiply fuel-cell electrical efficiency (nfc) by 0.97.
Y-E30	Multiply initial electric-storage inventory (Sees0) by 1.08.

TABLE XX
YANGSHAN PORT EQUIPMENT CONSTRAINT ADJUSTMENTS

Task	*Requested modifications*
Y-Q01	Add Pb[t5] ≤ 19.2 MW.
Y-Q02	Add Pesc[t6] ≤ 4.8 MW.
Y-Q03	Add Pesd[t7] ≤ 4.8 MW.
Y-Q04	Add Sees[t8] ≤ 19.2 MWh.
Y-Q05	Add Ped[t9] ≤ 4.8 MW.
Y-Q06	Add Pfc[t10] ≤ 3.84 MW.
Y-Q07	Add Hb[t11] ≤ 19.2 MW H_2.
Y-Q08	Add Hhsc[t12] ≤ 2.88 MW H_2.
Y-Q09	Add Hhsd[t13] ≤ 2.88 MW H_2.
Y-Q10	Add Shes[t14] ≤ 9.6 MWh H_2.
Y-Q11	Add Pb[t2] ≤ 18.8 MW; Add Pb[t3] ≤ 18.8 MW; Add Pb[t4] ≤ 18.8 MW.
Y-Q12	Add Pesc[t6] ≤ 4.7 MW; Add Pesc[t13] ≤ 4.7 MW; Add Pesc[t20] ≤ 4.7 MW.
Y-Q13	Add Pesd[t21] ≤ 4.7 MW; Add Pesd[t22] ≤ 4.7 MW; Add Pesd[t23] ≤ 4.7 MW.
Y-Q14	Add Sees[t2] ≤ 18.8 MWh; Add Sees[t3] ≤ 18.8 MWh; Add Sees[t4] ≤ 18.8 MWh.
Y-Q15	Add Ped[t6] ≤ 4.7 MW; Add Ped[t13] ≤ 4.7 MW; Add Ped[t20] ≤ 4.7 MW.
Y-Q16	Add Pfc[t21] ≤ 3.76 MW; Add Pfc[t22] ≤ 3.76 MW; Add Pfc[t23] ≤ 3.76 MW.
Y-Q17	Add Hb[t2] ≤ 18.8 MW H_2; Add Hb[t3] ≤ 18.8 MW H_2; Add Hb[t4] ≤ 18.8 MW H_2.
Y-Q18	Add Hhsc[t6] ≤ 2.82 MW H_2; Add Hhsc[t13] ≤ 2.82 MW H_2; Add Hhsc[t20] ≤ 2.82 MW H_2.
Y-Q19	Add Hhsd[t21] ≤ 2.82 MW H_2; Add Hhsd[t22] ≤ 2.82 MW H_2; Add Hhsd[t23] ≤ 2.82 MW H_2.
Y-Q20	Add Shes[t2] ≤ 9.4 MWh H_2; Add Shes[t3] ≤ 9.4 MWh H_2; Add Shes[t4] ≤ 9.4 MWh H_2.
Y-Q21	Add Pb[t8] ≤ 19.4 MW; Add Pesc[t8] ≤ 4.85 MW.
Y-Q22	Add Pesd[t9] ≤ 4.85 MW; Add Sees[t9] ≤ 19.4 MWh.
Y-Q23	Add Ped[t10] ≤ 4.85 MW; Add Pfc[t10] ≤ 3.88 MW.
Y-Q24	Add Hb[t11] ≤ 19.4 MW H_2; Add Hhsc[t11] ≤ 2.91 MW H_2.
Y-Q25	Add Hhsd[t12] ≤ 2.91 MW H_2; Add Shes[t12] ≤ 9.7 MWh H_2.

Task	*Requested modifications*
Y-Q26	Add Pb[t9] + Pb[t10] ≤ 37 MW.
Y-Q27	Add Pesc[t7] + Pesc[t19] ≤ 9.25 MW.
Y-Q28	Add Pesd[t9] + Pesd[t10] ≤ 9.25 MW.
Y-Q29	Add Sees[t7] + Sees[t19] ≤ 37 MWh.
Y-Q30	Add Ped[t9] + Ped[t10] ≤ 9.25 MW.

TABLE XXI
PORT LOGISTICS ADJUSTMENTS FOR YANGSHAN PORT

Task	*Requested modifications*
Y-L01	Increase vessel v2's earliest arrival by 2 h.
Y-L02	Decrease vessel v2's earliest arrival by 1 h.
Y-L03	Increase vessel v2's container workload by 20 containers.
Y-L04	Multiply vessel v2's shore-power demand by 0.92.
Y-L05	Increase vessel v2's latest departure by 2 h.
Y-L06	Increase earliest arrival for vessels v2 and v5 by 1 h.
Y-L07	Multiply vessel v2's shore-power demand by 1.07; Multiply vessel v5's shore-power demand by 0.95.
Y-L08	Increase vessel v2's earliest arrival by 1 h; Increase vessel v2's container workload by 15 containers.
Y-L09	Increase vessel v2's earliest arrival by 1 h; Multiply vessel v5's shore-power demand by 1.06.
Y-L10	Increase available QC count (NqcCount) by 1 unit; Increase vessel v2's container workload by 10 containers.
Y-L11	Increase vessel v3's earliest arrival by 2 h.
Y-L12	Decrease vessel v3's earliest arrival by 1 h.
Y-L13	Increase vessel v3's container workload by 20 containers.
Y-L14	Multiply vessel v3's shore-power demand by 0.92.
Y-L15	Increase vessel v3's latest departure by 2 h.
Y-L16	Increase earliest arrival for vessels v3 and v6 by 1 h.
Y-L17	Multiply vessel v3's shore-power demand by 1.07; Multiply vessel v6's shore-power demand by 0.95.
Y-L18	Increase vessel v3's earliest arrival by 1 h; Increase vessel v3's container workload by 15 containers.
Y-L19	Increase vessel v3's earliest arrival by 1 h; Multiply vessel v6's shore-power demand by 1.06.
Y-L20	Increase available YC count (NycCount) by 1 unit; Increase vessel v3's container workload by 10 containers.
Y-L21	Increase vessel v4's earliest arrival by 2 h.
Y-L22	Decrease vessel v4's earliest arrival by 1 h.
Y-L23	Increase vessel v4's container workload by 20 containers.
Y-L24	Multiply vessel v4's shore-power demand by 0.92.
Y-L25	Increase vessel v4's latest departure by 2 h.
Y-L26	Increase earliest arrival for vessels v4 and v7 by 1 h.
Y-L27	Multiply vessel v4's shore-power demand by 1.07; Multiply vessel v7's shore-power demand by 0.95.
Y-L28	Increase vessel v4's earliest arrival by 1 h; Increase vessel v4's container workload by 15 containers.
Y-L29	Increase vessel v4's earliest arrival by 1 h; Multiply vessel v7's shore-power demand by 1.06.
Y-L30	Increase available ECT count (NectCount) by 1 unit; Increase vessel v4's container workload by 10 containers.

TABLE XXII
YANGSHAN PORT ENERGY LOGISTICS COUPLED ADJUSTMENTS

Task	*Requested modifications*
Y-X01	Multiply grid electricity purchase tariff at t4, t5 by 1.07; Increase vessel v2's earliest arrival by 1 h.
Y-X02	Multiply grid electricity purchase tariff at t5, t6 by 1.07; Increase vessel v3's earliest arrival by 1 h.
Y-X03	Multiply grid electricity purchase tariff at t6, t7 by 1.07; Increase vessel v4's earliest arrival by 1 h.
Y-X04	Multiply available power of res at t7, t8 by 0.95; Multiply vessel v2's shore-power demand by 1.05.
Y-X05	Multiply available power of res at t8, t9 by 0.95; Multiply vessel v3's shore-power demand by 1.05.
Y-X06	Multiply available power of res at t9, t10 by 0.95; Multiply vessel v4's shore-power demand by 1.05.
Y-X07	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v2's container workload by 10 containers.
Y-X08	Multiply maximum electric-storage inventory (Seesmax) by 1.09; Increase vessel v3's container workload by 10 containers.
Y-X09	Multiply external hydrogen purchase price at t12, t13 by 1.06; Increase vessel v4's earliest arrival by 1 h; Multiply vessel v4's shore-power demand by 1.03.
Y-X10	Multiply external hydrogen purchase price at t13, t14 by 1.06; Increase vessel v2's earliest arrival by 1 h; Multiply vessel v2's shore-power demand by 1.03.

III. Evaluation Criteria

Coder and Interpreter use five dimensions scored at 0, 10 or 20 points, totaling 100. Coder implementation validity uses only 0 or 20. Scores combine programmatic checks and LLM-assisted grading.

Coder groups are Full, No-MD and No-FS; Interpreter groups are Full, No-NC and No-EO. Each group has 300 samples, with one answer per task. Group scores are sample means.

A. Coder scoring criteria

TABLE XXIII
Detailed Coder rubric

Dimension	20 points	10 points	0 points
Modification targets	All requested model objects, data fields, devices or variables are correctly selected.	Some requested targets are correct, but a target is omitted or locally incorrect.	The principal object or field is incorrect, or the requested modification is absent.
Numerical values and ranges	All requested values, multipliers, signs, periods and vessel/resource indices are correct.	A subset is correct, with a local value, index or coverage error.	A principal value, bound or range is wrong, or the main numerical requirement is not represented.
Requirement completeness	Every requested operation or constraint is expressed, including all parts of a coupled task.	Some requirements are satisfied but one or more operations or constraint parts are missing.	The main requirement is not implemented, or most required operations are missing.
Modification boundaries	Changes are confined to the requested inputs or newly added constraints; the original objective and other model components are preserved.	A limited extra change occurs while the main model and objective remain intact.	The original objective/constraints are improperly deleted or deactivated, or material unrelated changes are introduced.
Implementation validity	The candidate passes its required checks, restricted execution, data validation and model-construction checks.	Not used for this dimension.	Any required check, execution, data validation or model-construction step fails.

B. Interpreter scoring criteria

TABLE XXIV
Detailed Interpreter rubric

Dimension	20 points	10 points	0 points
Correctness of key conclusions	Total-cost direction, both focus-indicator directions, solver status and the principal answer agree with the independent reference.	The principal answer is correct, with a local conclusion or focus-indicator error.	The principal direction or answer is wrong, or the requested main analysis is absent.
Numerical correctness	All reported values match the reference results or correctly reconstructed values, allowing for rounding.	A secondary local numerical error occurs while the principal quantitative comparison remains valid.	A principal cost, difference or percentage is wrong, or widespread indicator errors invalidate the main quantitative comparison.
Factual completeness	All required cost and indicator summaries, total-cost comparison, two-indicator comparison, cost-source explanation and scope assessment are covered.	Some required content is missing, but the main requirements are answered.	The main requirements are unanswered or large portions of the required summaries or explanation are absent.
Evidence support	The main explanations of cost increases/decreases, indicator relationships and scope are correct and supported by supplied or correctly reconstructed facts.	Some evidence or a local relationship is insufficient or incorrect, without invalidating the principal explanation.	The principal explanation is absent, contradicts the independent evidence or makes an unsupported causal attribution.
Applicability and uncertainty	Statements respect the active case, variable meanings, totals, peaks, final states, raw-record coverage and model-based scope.	A local statement mildly exceeds the evidence or leaves a limited scope issue unclear.	A major claim exceeds the supplied evidence, invents a key device/result or guarantees a real-world outcome.

C. Safeguard evaluation

The static test set contains 150 compliant and 150 noncompliant snippets, with 50 of each per port. Table XXV summarizes the noncompliant categories.

TABLE XXV
COMPOSITION OF THE 150 NONCOMPLIANT SNIPPETS

Noncompliant category	Examples of prohibited operations	Number
External access	Imports, direct file writing, data-frame file export or other disallowed external/output calls.	30
Dynamic execution and reflection	eval, exec, compile, getattr or dunder attribute access.	30
Original model mutation	Deactivate the objective, fix original variables, alter original bounds, delete components or directly rewrite original model data.	30
Stage or object violations	Mix incompatible modification stages, create unauthorized components, access inactive objects or use disallowed data accessors.	30
Unsupported syntax	while, function definitions, try/except, with or lambda constructs outside the accepted code contract.	30

Safeguard Score assigns equal weight to the allowance and blocking rates. No-SG allows all snippets and receives a score of 50.

D. Pass and Accuracy

$$Pass = \frac{\sum_{i=1}^{300} I_{pass}(i)}{300}$$

I_pass(i) is 1 when the candidate passes the required checks, restricted execution and data validation, and completes model construction; it is 0 in all other cases.

$$Accuracy = \frac{\sum_{i=1}^{300} I_{correct}(i)}{300}$$

I_correct(i) is 1 when the candidate correctly implements every requested modification without unintended changes; it is 0 in all other cases. Failed candidates are also scored 0.

All proportion metrics lie between 0 and 1. Category-specific Accuracy uses 90 samples each for energy, equipment and logistics, and 30 for coupled tasks.

IV. End to End Interaction Examples

The examples show the user request, generated code, checks and results for each port.

TABLE XXVI
End to end example for Zhoushan Port

Zhoushan Port Electricity Price Change

User input

Increase the electricity purchase price in t18–t20 by 50%, with all other inputs unchanged. Do purchases move to other periods?

Coder generated modification

The selected context contains electricity prices and data-edit examples. The code is added before model construction.

```
for _t in ("t18", "t19", "t20"):
    scenario_time_data.loc[_t, "Ceb"] *= 1.5
```

Safeguard validation and execution

Human approval and AST checks passed. Only Ceb in t18–t20 changed, from 86.75 to 130.13 USD/MWh. Existing constraints were retained. Both solutions were optimal.

Interpreter result summary

Indicator	Baseline	What-if
Purchases in t18–t20 (MWh)	50.15	28.16
Purchases outside t18–t20 (MWh)	278.48	300.47
Total purchases (MWh)	328.63	328.63
Total vessel waiting (h)	12	19
Total cost (USD)	20,642.00	22,070.78

Purchases decrease by 21.99 MWh in t18–t20 and increase by the same amount in other periods. Total purchases are unchanged. Vessel waiting increases by 7 h, and total cost increases by 6.92%.

TABLE XXVII
END TO END EXAMPLE FOR THE PORT OF HOUSTON

Port of Houston Vessel Arrival Adjustment

User input

Delay vessel v3's expected arrival by 1 h. Keep all other vessel parameters unchanged.

Coder generated modification

The selected context contains Houston case details, vessel arrival data and data-edit examples. The code is added before model construction.

```
scenario_vessel_data.loc["v3", "ak"] += 1
```

Safeguard validation and execution

Human approval and AST checks passed. Only v3's arrival parameter ak changed, from 4 to 5 h. Other vessel parameters were retained. Both solutions were optimal.

Interpreter result summary

Indicator	Baseline	What-if
v3 expected arrival (h)	4	5
v3 berth start (period)	4	5
v3 berth end (period)	11	12
v3 waiting time (h)	0	0
Total cost (USD)	121,610.72	121,689.96

Vessel v3 moves from t4–t11 to t5–t12 at berth b4. Its waiting time remains zero. Electricity purchase cost decreases by USD 97.74, while natural-gas cost increases by USD 176.98. Total cost increases by USD 79.25 (0.0652%).

TABLE XXVIII
END TO END EXAMPLE FOR YANGSHAN PORT

Yangshan Port Grid Purchase Power Limits

User input

Limit grid purchases to 12 MW in t3 and t4 only. Retain all existing constraints.

Coder generated modification

The selected context contains the grid-purchase variable Pb and constraint-addition examples. The code is added after model construction.

```
model.GridPurchaseCap_t3 = pyo.Constraint(
    expr=model.Pb["t3"] <= 12
)

model.GridPurchaseCap_t4 = pyo.Constraint(
    expr=model.Pb["t4"] <= 12
)
```

Safeguard validation and execution

Human approval and AST checks passed. The new limits apply only to Pb in t3 and t4. Existing constraints were retained. Both solutions were optimal.

Interpreter result summary

Indicator	Baseline	What-if
Grid purchases in t3 (MW)	13.2415	12.0000
Grid purchases in t4 (MW)	15.9748	12.0000
Total grid purchases (MWh)	90.41	89.20
External hydrogen purchases (MWh H_2)	51.83	52.63
Total cost (USD)	6,836.48	6,885.35

Both limits are binding. Electrolyzer electricity use decreases by 1.33 MWh, while external hydrogen purchases increase by 0.80 MWh. Total cost increases by USD 48.87 (0.7148%), including USD 8.81 for electricity and USD 40.05 for hydrogen. Container throughput and total vessel waiting are unchanged.

Values are rounded for display. Differences and percentages use unrounded results.

V. SCOPE AND LIMITATIONS

The tests cover three deterministic port cases. Pass measures model construction, and Interpreter is evaluated with verified reference scenarios. The three end-to-end examples illustrate the workflow but do not establish an overall success rate. Results may vary with LLM outputs and have not been tested under forecast errors, real-time disturbances or other port configurations.